

Robin Lange

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Robotics and Autonomous Systems | Perception, Mission Logic and Physical Testing

Education

Georgia Southern University

Aug. 2023–May 2027

B.S. Mechanical Engineering, Vehicle Design & Testing concentration

GPA: 4.00/4.00

President's List; Honors College Scholar; Tau Beta Pi (junior upper eighth); Allen E. Paulson Scholar.

Descartes-Gymnasium Neuburg an der Donau

Sept. 2014–June 2023

Allgemeine Hochschulreife / General Higher Education Entrance Qualification

Germany

Honors Scholar; top 10% of graduating class. Final German grade: 1.2 (1.0 highest); 791/900 points.

Research Experience

Institute for Human & Machine Cognition (IHMC)

May–Aug. 2026

National Center for Collaborative Autonomy (NCCA)

Collaborative Robotics Autonomy Intern

Pensacola, FL

- Built a Rust/Python collaborative-autonomy codebase for two Boston Dynamics Spot robots; implemented pickup/drop-off sequencing and Joint Activity Graph coordination for group decisions.
- Combined inverse kinematics for coarse approach with local perception for final grasping, enabling hardware pickups despite tracking drift and imperfect perception.
- Implemented coordinate transforms and topic-based Zenoh communication so agents with different capabilities could share task state; a successor later added a drone.
- Integrated Motive motion capture and MCAP/Foxglove logging; instrumented a manually operated Husky AGV as a mobile, raised drop-off platform.

Intelligent Vehicles Navigation Research Laboratory

Oct. 2023–present

Georgia Southern University

Undergraduate Researcher

Statesboro, GA

- Integrated visual SLAM, nvblox mapping, Nav2 and behavior-tree mission logic on Jetson/RealSense UAV hardware; evaluated localization in a 110-second cart trial (14.5 cm RMSE on valid reference segments).
- Developed a LiDAR hand-motion interface for UAV control; coauthored two published SAE papers on lane perception and vehicle-simulation calibration.
- Honors thesis: commissioning a PD longitudinal-control benchmark in MATLAB and AVL; planning matched physical, always-on learned and reliability-gated preview comparisons under the same safety-supervised MPC.

CARISSMA Institute of Safety in Future Mobility (C-ISAFE)

July 1–Aug. 15, 2025

Technische Hochschule Ingolstadt

Vehicle Testing Research Intern

Ingolstadt, Germany

- Investigated how a soft truck-rear target appeared to radar, LiDAR and camera sensors; supported vehicle measurements and analyzed qualitative LiDAR/4D-radar returns with team scripts.
- Used the sensor observations to refine the target's tarpaulin covering and geometry; built its quick-assembly and fold-away mechanisms from an earlier CAD/foam concept.

Fraunhofer Institute for Transportation and Infrastructure Systems IVI

June 1–Aug. 11, 2024

Advanced Air Mobility research

Autonomous Systems and AI Intern

Ingolstadt, Germany

- Developed “Mission AI” decision and execution logic in Python, ROS 2 and py_trees; compared state-machine and behavior-tree approaches within an existing autonomy framework.
- Tested flight-phase transitions and emergency interruption in simulation, evaluated resource use and documented results for the research team.

Technical Skills

Programming: Python, Rust, MATLAB, C++; functional/procedural programming; entity-component-system (ECS) patterns.

Robotics: ROS 2/DDS, Zenoh, Protocol Buffers, Nav2; inverse kinematics, behavior trees, coordinate transforms, visual SLAM.

Hardware and tools: Boston Dynamics Spot, TurtleBot 4/Create 3, motion capture, LiDAR, RGB-D cameras, Jetson, Gazebo, SolidWorks, Git/Linux, MCAP/Foxglove.

Selected Awards and Scholarships

Third Place Undergraduate, Gulfstream Innovation, Engineering and Flight Excellence Awards – April 24, 2026. *Development of a Pilot AI for Autonomous UAV Inspection in Industry 5.0 Using Visual SLAM and Behavior Trees*; CEC Research Symposium.

Second Place Undergraduate, Georgia Power Innovation Awards – April 25, 2025. *Drone Guiding Through Equivalent Hand Movement Transversion*; CEC Research Symposium.

NCUR Engineering Division Video Contest Awardee – Council on Undergraduate Research, 2026.

Honors College Continuing Scholarship (2024–27; fewer than 15% of Honors College students receive college scholarships); Charles Robert Pound, Sr. Scholarship (2026–27); Paulson Scholar (2025); Vickie Hawkins Scholarship (2024).

Published Research

Soloiu, V., Sutton, T., Mehrzed, S., **Lange, R.**, Zimmerman, C., and Peralta Lopez, G. “Data-Driven Calibration of Vehicle Simulations for Accurate Control Response in Development and Testing.” *SAE Technical Paper 2026-01-0058*, 2026. doi:10.4271/2026-01-0058.

Soloiu, V., Mehrzed, S., Kroeger, L., Pierce, K., Sutton, T., and **Lange, R.** “Development of a Rule-Based Lidar Lane-Keeping Assistance System for Autonomous Vehicles Applications.” *SAE Technical Paper 2025-01-8025*, 2025. doi:10.4271/2025-01-8025.

Preprint and Manuscript in Preparation

Soloiu, V., Sutton, T., **Lange, R.**, and Zimmerman, C. “Warehouse Platooning Framework Using a Sensor-Rich Leader and Minimal Perception Follower.” SSRN 7067358. Preprint; not peer reviewed.

Lange, R., Sezgin, F., Inderst, M., Sutton, T., Lugner, R., Brandmeier, T., and Soloiu, V. “Design and Validation of a Soft Crash Target for Multimodal Sensor-Based Truck Underride Testing.” Manuscript in preparation.

Lange, R. Honors thesis in progress: reliability-gated lead-vehicle prediction for safety-supervised model predictive ACC. PD benchmark commissioning is underway; matched physical and learned-preview comparisons are planned.

Additional Research Presentations

Southern Regional Honors Council Conference, 2026, Montgomery, AL: Intelligent UAV Navigation with V-SLAM and Behavior Trees for Industry 5.0 Applications.

Southern Regional Honors Council Conference, March 2025, Chattanooga, TN: 3D-Space Hand Trajectory Tracking for Intuitive UAV Control.

Teaching and Academic Preparation

Physics I/II teaching assistant (Jan. 2025–present): evaluate experimental reports.

Computing for Engineers laboratory assistant (Jan.–May 2024): guided MATLAB and programming labs.

First-Year Experience peer leader (Aug.–Dec. 2024).

Selected completed coursework: Calculus I–III, Differential Equations, Numerical Methods, Dynamics, Mechatronics, Circuit Analysis, Program Design, Logic Circuit Design, Measurement & Instrumentation, Vehicle Design & Testing, Mechanism Design, Machine Design.

Languages: German (native), English (fluent), French (DELF B1).